Abdul Khalad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.28
Jawaharlal Nehru Govt Polytechnic	Class XII (State Board Of Technical Education & Training)	2016	86.13%
Good Shepherd High School	Class X (Board of Secondary Education Andhra Pradesh)	2013	9.70%

PROFESSIONAL EXPERIENCE

International Advanced Research Centre for Powder Metallurgy and New Materials (ARCI)

Project Technical Assistant

- Development of Indigenous Lithium Titanate and Lithium Iron Phosphate electrode materials in a High-energy ZOZ mill unit for the manufacturing of lithium-ion batteries.
- Process development and testing of supercapacitors in a pilot plant.
- Optimization of multi-tab welding process for lithium-ion battery.

RESEARCH EXPERIENCE

Indian Institute of Technology Kanpur

Research Intern - Micromachining Lab

- Two-dimensional analysis of Wire Electrochemical Turning and Wire Electrochemical Slitting processes using COMSOL Multiphysics software.
- Reduced the overcut by coating non-conductive material on the electrode.
- Simulate the effect in COMSOL Multiphysics and validate with the experimental result.

Defence Research and Development Lab, Hyderabad

- Fabrication of the inboard panel of a wing using High-speed Vertical CNC milling.
- Developed a part program using Master CAM software and performed simulations of the tool-work interface

PROJECTS

Numerical & Experimental investigation of heat transfer enhancement in a two-pass double-pass pipe heat exchanger with longitudinal strips

- Fabrication of longitudinal strip inserts for a double-pipe heat exchanger and heat transfer enhancement was explored.
- Performed numerical simulations using commercial CFD software and validated the experimental results

Automatic Side Stand Retrieval System

- Design and Fabrication of sprocket, pinion, and lever for a two-wheeler in resting and riding condition

TECHNICAL SKILLS

- AutoCAD Mechanical: Design and drafting software
- SolidWorks: 3D CAD Modelling software

- Abaqus: Modelling and simulation
- Ansys Fluent: CFD Commercial Software
- Languages: MATLAB, Python

INTERNATIONAL CONFERENCE PARTICIPATION

- Oral presentation at the 11th International Conference on Multiscale Materials Modeling 2024, in Prague (Czech Republic).
- Oral presentation at the CAMS 2024 Advanced Materials and Manufacturing, in Adelaide (Australia).
- Poster presentation at the 4th Asia-Pacific International Conference on Additive Manufacturing (APICAM), in Melbourne (Australia), July 2025.
- Oral presentation at the Materials Science & Technology (MS&T) 2025, in Columbus, Ohio (USA).

PUBLICATIONS

1. **A. Khalad**, G. Telesang, K. Kadali, P.N. Zhang, W. Xu, and V. Chinthapenta, "A Generalized Machine Learning Framework for Data-Driven Prediction of Relative Density in Laser Powder Bed Fusion Parts," International Journal of Advanced Manufacturing, doi: 10.1007/s00170-024- 14735-wTechnology, (Q1, Accepted, IF = 3.1).
2. **A. Khalad***, A. Anjum*, S. S. Akhtar, and A. A. Shaik, "A meta-heuristic evolutionary algorithm combined with XGBoost to predict the geometry characteristics of laser-based micro-milling in PMMA-based microchannels" submitted to the Optics and Laser technology. (Q1, Accepted, IF = 5, *Equal authorship).
3. A. Maqbool*, **A. Khalad***, and N. Z. Khan, "Prediction of corrosion rate for friction stir processed WE43 alloy by combining PSO-based virtual sample generation and machine learning," Journal of Magnesium and Alloys, vol. 12, no. 4, pp. 1518-1528, Apr. 2024, doi: 10.1016/j.jma.2024.04.012. (Q1, Accepted, IF = 13.8, * = Equal authorship)
4. A. Hamza, **A. Khalad**, and D. S. Kumar, "Enhanced production of mycelium biomass and exopolysaccharides of Pleurotus ostreatus by integrating response surface methodology and artificial neural network," Bioresource Technology, vol. 399, p. 130577, May 2024, doi: 10.1016/j.biortech.2024.130577. (Q1, Accepted, IF = 9)
5. J. E. Manikanta, **A. Khalad**, S. Santosh, N. K. Gurajala, and C. Prasad, "Employing artificial neural networks and gray relational analysis to modeling and optimization for sustainable machining process applications," submitted to the International Journal on Interactive Design and Manufacturing (Q2, Accepted, IF = 2.5).
6. M. P. Shankar, A. Hamza, **A. Khalad**, G. Shanthi, S. Kuppireddy, and D. S. Kumar, "Engineering Mushroom Mycelium for a Greener Built Environment: Advancements in Mycelium-based Biocomposites and Bioleather", Food Biosci., vol. 62, p. 105577, Dec. 2024, doi: 10.1016/j.fbio.2024.105577. (Q1, Accepted, IF = 5.9).
7. P. A. Imtiyaz, **A. Khalad**, and S. Abdul Muqtadir, "Prediction of thermal conduction of energy piles using various machine learning models," International Journal of Geotechnical Engineering, vol. 0, no. 0, pp. 1-13, 2024, doi: 10.1080/19386362.2024.2362024. (Q2, Accepted, IF = 1.3)